

Combinatorics PhD Exam
May 20, 1991

Do 9 out of 12 problems. Show all of your work.

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1. How many k -tuples $(A_1, A_2, \dots, A_k)$ of nested subsets of an n -set are there, where $A_1 \subseteq A_2 \subseteq \dots \subseteq A_k$.
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2. How many words of length n are there in the alphabet (a, b, c) with no two adjacent a 's? With no two adjacent letters equal?
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3. Find the generating function for the sequence $\{a_n\}$ given by the recurrence

$$a_n = a_{n-1} + 6a_{n-2} + 2^n$$

where $a_0 = 1$ and $a_1 = 3$.

4. Bipartite graphs:

(a) Determine for which values of m and n the complete bipartite graph K_{mn} is (1) planar; (2) Eulerian; (3) Hamiltonian.

(b) Prove: A plane graph where every face has an even number of edges must be bipartite.

5. Prove that $\sum_{k=1}^n S(n, k)x(x-1)\dots(x-k+1) = x^n$.
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6. Use Möbius inversion to find the number of integers between 1 and n and coprime to n , given the prime factorization of n .
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7. Prove or disprove: There is a matching from V to W in a bipartite graph if, for some fixed k , there are k or more edges incident with each vertex of V and k or fewer edges incident with each vertex of W .

8. Let $A_{m,n}$ be the collection of all sets $X = \{N_1, N_2, \dots, N_k\}$ where k is a non-negative integer, and for each $i \leq k$, N_i is a positive integer divisor of $N = 2^m 3^n$, and where $N_i | N_j$ only if $i = j$. Define a partial order on $A_{m,n}$, where $X = \{N_1, N_2, \dots, N_k\}$, $X' = \{N'_1, N'_2, \dots, N'_l\}$, by $X \leq X'$ if and only if $N_i \in X \Rightarrow$ there exists $N'_j \in X'$ with $N_i | N'_j$. Prove that $A_{m,n}$ is a distributive lattice, and determine its poset of join-irreducible elements.
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9. Let I be a collection of subsets of E , where E is finite, and suppose that $I \neq \emptyset$ and that $Y \in I$, $X \subseteq Y \Rightarrow X \in I$. Let $c : E \rightarrow \mathbb{R}^+$ assign a positive real number to each element of E . Consider the optimization problem:

$$\text{Find } \max_{X \in I} \sum_{x \in X} c(x).$$

Prove that the greedy algorithm solves this problem if and only if I is the collection of independent sets of some matroid on E .

10. State the MacWilliams identity, and use it to compute the weight polynomial of the (15,11) Hamming binary code.
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11. A (v, k, λ) -design Σ is an incidence structure consisting of a set S of v points and a collection of k -subsets of S called blocks, such that every pair of points of S is contained in precisely λ blocks.

Prove that, in the range $30 \leq v \leq 45$, a $(v, 6, 1)$ -design exists if and only if $v = 31$.

12. Use the Hall Multiplier Theorem

- (i) to prove the non-existence of a cyclic (31, 10, 3) difference set.
- (ii) to find a cyclic (31, 6, 1) difference set.